

Foundation (Jalapeno)

Q1. Solve for x : $x + 5 = 11$

- (A) $x = 6$
- (B) $x = 7$
- (C) $x = 8$
- (D) $x = 9$
- (E) $x = 10$

Q2. A spaceship travels $3x - 2$ miles in x hours. If it travels 7 miles, how many hours did it travel?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q3. Solve for y : $2y - 4 = 8$

- (A) $y = 3$
- (B) $y = 4$
- (C) $y = 5$
- (D) $y = 6$
- (E) $y = 7$

Q4. A lab experiment requires a temperature of $2x + 5$ degrees. If the temperature is 11 degrees, what is the value of x ?

- (A) $x = 1$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 4$
- (E) $x = 5$

Q5. Solve for z : $z - 2 = 9$

- (A) $z = 7$
- (B) $z = 8$
- (C) $z = 9$
- (D) $z = 10$
- (E) $z = 11$

Q6. The height of a rocket is $5x - 3$ feet after x seconds. If the height is 22 feet, how many seconds have passed?

- (A) $x = 3$
- (B) $x = 4$
- (C) $x = 5$
- (D) $x = 6$
- (E) $x = 7$

Q7. Solve for w : $w + 2 = 6$

- (A) $w = 2$
- (B) $w = 3$
- (C) $w = 4$
- (D) $w = 5$
- (E) $w = 6$

Q8. A planet's distance from the sun is $4x + 1$ million miles. If the distance is 25 million miles, what is the value of x ?

- (A) $x = 4$
- (B) $x = 5$
- (C) $x = 6$
- (D) $x = 7$
- (E) $x = 8$

Open Ended Questions (FRQ)

Q9. Solve the equation $2x + 5 = 11$ and check your solution for accuracy. Show all work.

Q10. A scientist mixes $3x - 2$ ounces of a chemical with $2x + 1$ ounces of water. If the total mixture is 15 ounces, solve for x and check your solution. Show all work.

Chef's Tips

- Emphasize the importance of checking solutions for accuracy
- Encourage students to use algebraic properties to simplify equations
- Remind students to follow the order of operations when solving multi-step equations